COMPSCI 689
Lecture 26: Course Wrap-Up

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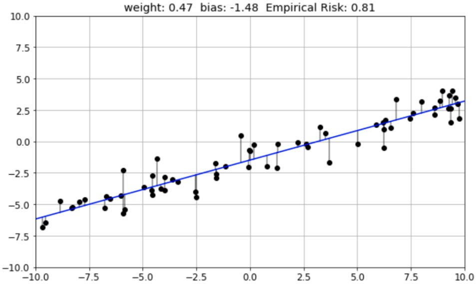
OLS Linear Regression (1800's)



Adrien-Marie Legendre

1805: Linear least squares method

Carl Friedrich Gauss

1822: Linear least squares optimality

Text to Images and Video (2024)

OpenAI

Google Research

Google Brain

What have we covered?

- **Tasks:** Regression, Classification, Clustering, Dimensionality Reduction, Density Estimation, Denoising [L1]
- **Foundations:** Numerical optimization [L2], multivariate probability [L4]
- **Learning Frameworks:** ERM, RRM, MLE, Variational Learning [L2, L7]
- **Supervised Models:** OLS, SVM/SVR, MLPs, CNNs, RNNs, Logistic Regression, Probabilistic Supervised Models [L3]
- **Unsupervised Models:** Mixture Models, Factor Analysis, Autoencoders, VAEs, Transformers, Diffusion [L3]

What have we covered?

- **Implementing custom models:** Homework 3 and 4 [L4]
- **Meta-Issues:** Generalization assessment, hyper-parameter selection, experiment design [L5,L6], computational efficiency, numerical stability [L5,L9]
- **Skills:** Graphing and results visualization [L8]
- **Tools:** NumPy, SciPy, PyTorch [L10]

Possible Final Exam Topics

- **Learning Frameworks:** ERM, MLE (including NLML), reconstruction-based
- **Supervised Models:** MLPs, RNNs, CNNs, classical and generalized probabilistic supervised models (regression, classification, linear, non-linear, etc.)
- **Unsupervised Models:** Multivariate normals, discrete and continuous product of marginals, mixtures, factor analysis, non-linear factor analysis/VAEs, autoencoders, transformers, text-to-X models
- **Probabilistic Inference:** Marginalization, conditioning, making predictions, handling missing data

Possible Final Exam Topics

- **Meta Issues:** Generalization assessment, hyper-parameter selection, experiment design, computational complexity, numerical stability, optimization convergence
- **Skills and Tools:** Code vectorization, recognizing/correcting problems with implementations (Numpy/Torch), interpreting experiment results and graphs

Exam Question Types

The focus of assessment is on *generalization* not *memorization*.

- **Explanation:** Explaining concepts or the relationships between concepts.
- **Analysis:** Analyzing derivations, equations, code, graphs, etc. to identify problems, suggest alternatives, select an optimal approach, etc.
- **Application/Synthesis:** Applying concepts and frameworks to derive equations, construct models etc.
- **Proofs:** Proving properties of models or other structures.

SAT/Fail Grades

- The UMass grad program allows students to choose a Satisfactory/Unsatisfactory (e.g., SAT/Fail) grading basis.
- In this class, SAT/fail grading can be requested via Canvas Inbox message with the subject “SAT request" by 11:59pm on the last day of classes.
- SAT grading requests can not be conditional (e.g., we do not accept requests of the form “If my grade is less than A I want a SAT, otherwise I want the A”).
- The minimum grade needed to earn a SAT is a C.
- SAT/fail grading can not be revoked after the deadline for submitting requests.
- If you are considering SAT/fail grading, make sure you understand the implications on your program requirements.

More Machine Learning

- **What if I have a small amount of training data?** Data augmentation. Bayesian machine learning.
- **What if I have a small amount of labeled data and a large amount of un-labeled data?** Semi-supervised learning. Active learning.
- **What if I have a small amount of data for the task I care about and a large amount of data for somewhat related tasks?** Domain adaptation, transfer learning.
- **What if I have data for several different, but related tasks?** Multi-task learning.

More Machine Learning

- **What if the data distribution I am trying to model is changing over time?** Lifelong learning, continual learning.
- **What if I'm concerned about robustness to arbitrary outliers during deployment?** Robust models, out-of-distribution detection.
- **What if I need to be robust to the possibility of direct attack against a deployed model?** Adversarial training.
- **What if my classification problem has severely imbalanced classes?** Instance weighted learning.

More Machine Learning

- **What if I have missing data in the inputs of my model?** Imputation, probabilistic modeling.
- **What if I need to predict outputs over a structure like a graph or a tree?** Structured prediction.
- **What if I need to predict a ranking over objects?** Learning to rank.
- **What if I need to understand what data variables are impacting model performance?** Feature selection.

More Machine Learning

- **What if I need to understand what training data cases are most impacting impacting model performance?** Influence functions.
- **What if I need to understand why my model makes the predictions it does?** Model explainability.
- **What if I want to make sure my model isn't prejudiced or racist?** Algorithmic bias detection and mitigation.
- **What if I need to reduce the deployment resource use of my model?** Model compression, pruning and distillation.
- **How do I keep up with advances in machine learning?**

Where to go from here? - ML Courses

- COMPSCI 651: Optimization
- COMPSCI 682: Deep Learning
- COMPSCI 687: Reinforcement Learning
- COMPSCI 688: Probabilistic Graphical Models

Where to go from here? - Applications Courses

- COMPSCI 685: Advanced Natural Language Processing
- COMPSCI 603: Robotics
- COMPSCI 646: Information Retrieval
- COMPSCI 670: Computer Vision
- COMPSCI 574/674: Intelligent Visual Computing

Where to go from here? - Ethics

- COMPSCI 508 - Ethical Considerations in Computing

The End

Thank You!

(Don't forget to complete your SRTI)